

Reading Support: Mathematics and Linguistic Structure

Companion to Week 3

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Mathematical description and computation answer different questions

Supports: A formal description is not automatically a mechanism

A mathematical description identifies objects satisfying stated conditions. A computation specifies transitions among configurations that produce or recognize an answer. Either can exist without the other being identified.

To call a description computational therefore adds a claim about effective steps, admissible inputs, and the conditions under which a process has completed.

MOLR §1.8

Physical computation requires interpretation at both ends

Supports: Abstract computation suppresses its physical encoding

An abstract machine manipulates formal configurations. A physical computer manipulates material states that count as those configurations only through encoding and decoding relations.

The abstract account deliberately ignores the material implementation. This makes different realizations comparable, but it does not establish that every physical process computes every abstract function to which its states can be mapped.

MOLR §1.8

Linguistic structure exceeds the visible string

Supports: Linear order does not determine dependency

In “I persuaded, promised, expected, or wanted her to study,” the words occur in much the same order, but the understood relations differ. With persuade, her is understood as the student. With promise, the matrix subject is normally understood that way.

The string consequently underdetermines the dependency that interpretation requires.

MOLR §1.9

Several judgments hide behind one star

Supports: Grammaticality, meaningfulness, and truth can diverge

Linguists use a star for expressions judged unacceptable under an intended reading and context. The source of the judgment may involve word order, selection, interpretation, discourse conditions, or combinations of them.

“Colorless green ideas sleep furiously” shows that recognizable grammatical organization can survive severe interpretive difficulty. Truth adds a further, independent relation to the world.

MOLR §§1.9–1.10

Grammar can be necessary without being sufficient

Supports: Grammar is an INUS condition

An INUS condition is an insufficient but necessary part of an unnecessary but sufficient condition. Grammar alone does not make a linguistic act successful. Within a package that does succeed, however, the grammatical contribution may be indispensable.

Other packages may achieve the activity through different linguistic forms, gestures, shared circumstances, or repairs. Necessity must therefore be stated relative to the sufficient package under discussion.

MOLR §1.10

Typology needs an invariant independent of one exponent

Supports: Typology compares variants through explicit invariants

Cross-linguistic comparison cannot begin by treating the overt form of one language as the universal category. Languages may realize a shared dependency through order, agreement, case, a resumptive element, or no single dedicated exponent.

The proposed invariant must be abstract enough to admit genuine variants and restrictive enough to make the comparison empirically informative.

MOLR §1.11

Relative clauses illustrate form variation under a shared relation

Supports: Relative clauses preserve a relation across formal variation

A restrictive relative clause connects two predications through a shared element, commonly restricting the reference of a nominal expression. One semantic representation is $\lambda x[P(x) \wedge Q(x)]$.

This abstraction does not require an English relative pronoun or an unpronounced gap. Those are possible language-particular means of realizing the relation.

MOLR §1.11

Model construction begins before formal calculation

Supports: A model is selected through several admissibility filters

Conceptual vocabulary determines which objects and relations the modeling language can express. The language then defines a space of formally admissible models. Further assumptions select candidates, and empirical work provisionally validates some of them.

Each stage can remove possibilities. No stage by itself guarantees that the surviving formal object corresponds to linguistic reality.

MOLR §1.12

Verification concerns the model; validation concerns the target

Supports: Verification and validation are different achievements

Verification asks whether the model follows from the definitions, rules, and calculations of the chosen modeling language. Validation asks whether its interpretation fits the phenomenon for which it was constructed.

A grammar may generate exactly the observed corpus and still fail to predict new judgments, acquisition patterns, or variation. Formal correctness is necessary for the model's claim, but the empirical correspondence remains open.

MOLR §1.12